

MATH2030J - Discrete Mathematics

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Lectures 2 & 3 & 4



上海交通大學

SHANGHAI JIAO TONG UNIVERSITY

Announcements

- ▶ Remember: attendance to lectures is mandatory.
- ▶ If you cannot attend regularly, let me know.
- ▶ Coursework groups: open for free signup on Canvas.
- ▶ Students who havent chosen their group by tomorrow will be automatically assigned.
- ▶ Discussion session sheet will be released tomorrow.

- ▶ Course rules, logistics, and schedule.
- ▶ Problems you can keep in the back of your mind, to see how your insight changes throughout the course.

Dramatis Personae

Meet a cast of intriguing problems!

- ▶ A Game of Hats *Error-correcting code*
- ▶ A Busy Postal Worker *Sorting algorithm*
- ▶ $i\mathcal{R}rational \ell^{i\mathcal{R}raTioNa\ell} = \text{rational?}$ *Non-constructive proof*
- ▶ An Infinite Hotel *Bijection, cardinality*
- ▶ Smallest positive integer not definable in under sixty letters. *Logic*
- ▶ The Blue-Eyed Islanders *Induction*
- ▶ Colour everywhere! *The pigeonhole principle*
- ▶ Chomp! *Dynamic programming*
- ▶ How many musical notes can there be? *Groups, but not quite*

Today's objectives

- ▶ Compound propositions
- ▶ Truth tables
- ▶ Conditional statements
- ▶ Tautological equivalence
- ▶ Rules of inference

Ch. 1. Logic and Proofs

- 1.1. Propositional Logic
- 1.2. Rules of Inference
- 1.3. Predicates and Quantifiers
- 1.4. Introduction to Formal Proofs
- 1.5. (Informal) Proof Techniques

- ▶ Runze Cai, MATH1160J Sp. 2024 course
- ▶ K. H. Rosen, Discrete Mathematics and its Applications
- ▶ J. Gallier, Discrete Mathematics
- ▶ Frank Pfenning, Constructive Logic course:
<https://www.cs.cmu.edu/~fp/courses/15317-f09/schedule.html>

1.1. Propositional Logic

Definition

A *proposition* or *statement* is a declarative sentence that is either **true** or **false**, but not both.

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Examples

- ▶ Washington, D.C., is the capital of the United States of America.
- ▶ Toronto is the capital of Canada.
- ▶ $1 + 1 = 2$.
- ▶ $2 + 2 = 3$.

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A *proposition* or *statement* is a declarative sentence that is either **true** or **false**, but not both.

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- ▶ $2 + 2 = 3$.

Non-examples

- ▶ What time is it?
- ▶ Read this carefully.
- ▶ $x + 1 = 2$.
- ▶ $x + y = z$.

Propositional/Logical Variables

Denoted by $p, q, r, \dots$, or $A, B, C, \dots$

Boolean variables: True/False

- ▶ True: $T, 1, \top, \text{true}$
- ▶ False: $F, 0, \perp, \text{false}$

We sometimes denote the booleans by $\mathbb{B} = \{\top, \perp\}$.

Compound propositions:

Compound propositions are formed from existing propositions using logical operators. Atomic propositions cannot be expressed in terms of simpler propositions.

Logical operators

$\mathbb{B} \rightarrow \mathbb{B}$

- ▶ $\neg$, negation/not.

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Connectives: $\mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$

- ▶ $\wedge$, and
- ▶ $\vee$, or (inclusive or)
- ▶ $\rightarrow$, implies
- ▶ $\leftrightarrow$, if and only if (iff)

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Currying: $\mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$, or $\mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}$

$$\begin{aligned}\wedge : \mathbb{B} \times \mathbb{B} &\longrightarrow \mathbb{B} \\ (p, q) &\longmapsto p \wedge q\end{aligned}$$

can also be seen as:

$$\begin{aligned}\wedge : \mathbb{B} &\longrightarrow \mathbb{B} \rightarrow \mathbb{B} \\ p &\longmapsto \Lambda_p : q \in \mathbb{B} \mapsto \Lambda_p(q) := p \wedge q.\end{aligned}$$

Conjunction and disjunction truth tables

AND, $\wedge : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$

p	q	$p \wedge q$
0	0	
0	1	
1	0	
1	1	

OR (inclusive or), $\vee : \mathbb{B} \times \mathbb{B} \rightarrow \mathbb{B}$

p	q	$p \vee q$
0	0	
0	1	
1	0	
1	1	

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Conjunction and disjunction truth tables

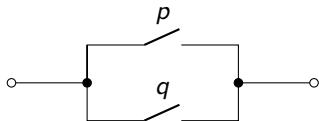
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Remark

- ▶ In the conjunction $p \wedge q$, the proposition p and q are called *conjuncts*;
- ▶ In the disjunction $p \vee q$, the proposition p and q are called *disjuncts*.

Negation, Exclusive OR, Bitwise Operations

NOT, $\neg : \mathbb{B} \rightarrow \mathbb{B}$

p	$\neg p$
0	
1	

XOR (exclusive or)

p	q	$p \oplus q$
0	0	
0	1	
1	0	
1	1	

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Bitwise OR, bitwise AND, bitwise XOR

We can extend those operations to bit strings of equal length:

01	1011	0110	
11	0001	1111	
11	1011	1111	bitwise OR
01	0001	0100	bitwise AND
10	1010	1011	bitwise XOR.

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Did you spot any mistakes?

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10	1010	1001	bitwise XOR.

Conditional Statements

Conditional statement (material implication)

- ▶ p : hypothesis/antecedent/premise
- ▶ q : thesis/conclusion/consequence

p	q	$p \rightarrow q$
0	0	1
0	1	1
1	0	0
1	1	1

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1	1	1

Equivalent forms

- ▶ if p , then q
- ▶ if p , q
- ▶ q if p
- ▶ q when p
- ▶ q unless $\neg p$
- ▶ p implies q
- ▶ p only if q
- ▶ p is sufficient for q
- ▶ q whenever p
- ▶ a necessary condition for p is q
- ▶ a sufficient condition for q is p
- ▶ q is necessary for p
- ▶ q follows from p

Conditional Statements

Conditional statement (material implication)

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0	0	1
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Equivalent forms

If (rain) Then (umbrella)
 p q

- ▶ if p , then q
- ▶ if p , q
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0 errors

There are ~~several errors~~ on this slide. Find them all.

Conditional-Disjunction Equivalence

Proposition

$p \rightarrow q$ and $\neg p \vee q$ have the same truth table.

Proof: Exercise.

~ If 0=1 then
I am the Pope. ~

p	q	$p \rightarrow q$	$\neg p$	$\neg p \vee q$
0	0	1	1	1
0	1	1	1	1
1	0	0	0	0
1	1	1	0	1

Conditional Statements

“No underage drinking”

If $\underbrace{\text{somebody}}_{A, B, C, \text{ or } D}$ is $\underbrace{\text{drinking alcohol}}_p$, then $\underbrace{\text{they are over 18 y/o.}}_q$.

Conditional Statements

“No underage drinking”

If $\underbrace{\text{somebody is drinking alcohol}}_{A, B, C, \text{ or } D}$, then $\underbrace{\text{they are over 18 y/o.}}_q$

p

- ▶ A is drinking beer
- ▶ B is drinking cola
- ▶ C is 55 y/o
- ▶ D is 15 y/o

Conditional Statements

“No underage drinking”

If somebody is drinking alcohol, then they are over 18 y/o.

$\underbrace{A, B, C, \text{ or } D}_{p} \quad \underbrace{\text{they are over 18 y/o.}}_q$

- ▶ A is drinking beer
- ▶ B is drinking cola
- ▶ C is 55 y/o
- ▶ D is 15 y/o

p	q	$p \rightarrow q$
0	0	1
0	1	1
1	0	0
1	1	1

Remark

We seek to find out whether a certain promise or guarantee is kept:

- ▶ either p is false
- ▶ or q is true

Conditional Statements

“No underage drinking”

If somebody is drinking alcohol, then they are over 18 y/o.

$A, B, C,$ or D

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Remark

We seek to find out whether a certain promise or guarantee is kept:

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p	q	$\neg p$	$\neg p \vee q$
0	0	1	1
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Converse/Inverse/Contrapositive/Negation

Given $p \rightarrow q$,

- ▶ Converse: $q \rightarrow p$
- ▶ Inverse: $\neg p \rightarrow \neg q$
- ▶ Contrapositive: $\neg q \rightarrow \neg p$
- ▶ Negation: $\neg(p \rightarrow q)$

p	q	$p \rightarrow q$	$\neg q \rightarrow \neg p$
0	0	1	1
0	1	1	1
1	0	0	0
1	1	1	1

Example: p -value

In null-hypothesis significance testing, the p -value is the probability of obtaining test results at least as extreme as the result actually observed, **under the assumption that the null hypothesis is correct.**

Exercise: fill the truth table

c

Given $p \rightarrow q$,

- ▶ Converse: $q \rightarrow p$
- ▶ Inverse: $\neg p \rightarrow \neg q$
- ▶ Contrapositive: $\neg q \rightarrow \neg p$
- ▶ Negation: $\neg(p \rightarrow q)$

$\neg p$	$\neg q$	p	q	$p \rightarrow q$	$\neg q \rightarrow \neg p$	$\neg p \rightarrow \neg q$	$\neg(p \rightarrow q)$
1	1	0	0	1	1	1	0
1	0	0	1	1	1	0	0
0	1	1	0	0	0	1	1
0	0	1	1	1	1	1	0

Biconditional Statements

if and only if (iff)

$$(p \rightarrow q) \wedge (q \rightarrow p)$$

p	q	$p \rightarrow q$	$q \rightarrow p$	$p \leftrightarrow q$
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	1	1	1

Remark:

Material implication $p \rightarrow q$ only means that in all cases when p is true, q is also true. For example, "If $1 + 1 = 3$, then I am the Pope" is true. So $p \rightarrow q$ **does not mean** that we can prove q from p .

Building Compound Propositions

Priority of connectives

(use brackets “(...)” when in doubt)

1. $\neg$

2. $\wedge$

3. $\vee$

4. $\rightarrow$

5. $\leftrightarrow$

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(use brackets “(...)” when in doubt)

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Conventions

(agreeing with LEAN4)

$$\blacktriangleright A \rightarrow B \rightarrow C = A \rightarrow (B \rightarrow C)$$

$$\blacktriangleright A \wedge B \wedge C = A \wedge (B \wedge C)$$

$$\blacktriangleright A \vee B \vee C = A \vee (B \vee C)$$

Building Compound Propositions

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Add parenthesis without altering the following propositions

$$(p \wedge (\neg q)) \leftrightarrow ((r \vee q) \rightarrow (p \rightarrow r))$$

$$((\neg(p \vee (q \wedge r))) \wedge (\neg r)) \vee (q \wedge p)$$

Tautology and Contradiction

A compound proposition that is...

- ▶ always true is a **Tautology**.

Truth table: all cases evaluates to 1. (e.g., $p \vee \neg p$)

p	$\neg p$	$p \vee \neg p$
1	0	1
0	1	1

- ▶ always false is a **Contradiction**.

Truth table: all cases evaluates to 0. (e.g., $p \wedge \neg p$)

p	$\neg p$	$p \wedge \neg p$
1	0	0
0	1	0

- ▶ neither of the above is a **Contingency**: "it depends".

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Truth table: all cases evaluates to 0. (e.g., $p \wedge \neg p$)

- ▶ neither of the above is a **Contingency**: “it depends”.

Equivalence

Propositions p and q are called **equivalent** iff $p \leftrightarrow q$ is a tautology.

We denote it by $p \leftrightarrow q$ (or $p \equiv q$).

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A compound proposition that is...

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- ▶ neither of the above is a **Contingency**: “it depends”.

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Examples

- ▶ $p \vee \neg p \leftrightarrow 1$

- ▶ $p \wedge \neg p \leftrightarrow 0$

- ▶ $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p) \leftrightarrow (\neg p \vee q)$

Tautological Equivalences

► Identity

$$p \vee 0 \Leftrightarrow p \quad (\max\{p, 0\} = p)$$

$$p \wedge 1 \Leftrightarrow p \quad (\min\{p, 1\} = p)$$

► Null

$$p \wedge 0 \Leftrightarrow 0 \quad (\min\{p, 0\} = 0)$$

$$p \vee 1 \Leftrightarrow 1 \quad (\max\{p, 1\} = 1)$$

► Idempotent

$$p \wedge p \Leftrightarrow p$$

$$p \vee p \Leftrightarrow p$$

► Absorption

$$p \wedge (p \vee q) \Leftrightarrow p$$

$$p \vee (p \wedge q) \Leftrightarrow p$$

Exercise:

Prove the equivalences by checking their truth tables.

p	0	$p \vee 0$	$(p \vee 0) \Leftrightarrow p$
1	0	1	1
0	0	0	1

p	q	$p \vee q$	$p \wedge (p \vee q)$	$(p \wedge p) \vee q$
1	1	1	1	1
1	0	1	1	1
0	1	1	0	1
0	0	0	0	0

- ▶ Double negation

$$\neg\neg p \Leftrightarrow p$$

Tautological Equivalence

- ▶ Double negation

$$\neg\neg p \Leftrightarrow p$$

- ▶ Negation laws

$$p \wedge \neg p \Leftrightarrow 0$$

$$p \vee \neg p \Leftrightarrow 1$$

Tautological Equivalence

► Double negation

$$\neg\neg p \Leftrightarrow p$$

► Negation laws

$$p \wedge \neg p \Leftrightarrow 0$$

$$p \vee \neg p \Leftrightarrow 1$$

► De Morgan's Laws

$$\neg(p \vee q) \Leftrightarrow (\neg p) \wedge (\neg q)$$

$$\neg(p \wedge q) \Leftrightarrow (\neg p) \vee (\neg q)$$

p	q	$\neg(p \vee q)$	$\neg p$	$\neg q$	$(\neg p) \wedge (\neg q)$
1	1	0	0	0	0
1	0	0	0	1	0
0	1	0	1	0	0
0	0	1	1	1	1

Tautological Equivalences

► Commutativity

$$p \wedge q \Leftrightarrow q \wedge p$$

$$p \vee q \Leftrightarrow q \vee p$$

$$(p \rightarrow q \rightarrow r) \Leftrightarrow p \rightarrow (q \rightarrow r)$$

► Associativity

$$(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$$

$$(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r)$$

$$(p \leftrightarrow q) \leftrightarrow r \Leftrightarrow p \leftrightarrow (q \leftrightarrow r)$$

$$(p \oplus q) \oplus r \Leftrightarrow p \oplus (q \oplus r)$$

$$(p \rightarrow q) \rightarrow r \not\Leftrightarrow p \rightarrow (q \rightarrow r)$$

► Distributivity

$$p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$$

AND
OR
IFF
XOR

Tautological Equivalence - Conditional Statements

- ▶ Conditional statements and compound propositions:

$$p \rightarrow q \Leftrightarrow \neg p \vee q$$

$$p \rightarrow q \Leftrightarrow \neg q \rightarrow \neg p$$

$$p \vee q \Leftrightarrow \neg p \rightarrow q$$

$$p \wedge q \Leftrightarrow \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \Leftrightarrow p \wedge \neg q$$

- ▶ Biconditional Statements:

$$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \Leftrightarrow \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \Leftrightarrow (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \Leftrightarrow p \leftrightarrow \neg q$$

- ▶ Cases

$$(p \rightarrow q) \wedge (p \rightarrow r) \Leftrightarrow p \rightarrow (q \wedge r)$$

$$(p \rightarrow q) \vee (p \rightarrow r) \Leftrightarrow p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \Leftrightarrow (p \vee q) \rightarrow r$$

$$(p \rightarrow r) \vee (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$$

Added Premise/Exportation

$$f(x, y) = z$$

► Added premise/exportation

$$(x, y) \mapsto z$$

$$(p \wedge q) \rightarrow r \Leftrightarrow p \rightarrow q \rightarrow r \Leftrightarrow p \rightarrow (q \rightarrow r)$$

$$x \mapsto f(x, \cdot) : y \mapsto f(x, y) = z$$

$$\Leftrightarrow q \rightarrow p \rightarrow r \Leftrightarrow q \rightarrow (p \rightarrow r)$$

$$y \mapsto f(\cdot, y) : x \mapsto f(x, y) = z$$

p	q	r	$p \wedge q$	r	$(p \wedge q) \rightarrow r$	p	$q \rightarrow r$	$p \rightarrow q \rightarrow r$
0	0	0	0	0	1	0	1	1
0	0	1	0	1	1	0	1	1
0	1	0	0	0	1	0	0	1
0	1	1	0	1	1	0	1	1
1	0	0	0	0	1	1	1	1
1	0	1	0	1	1	1	1	1
1	1	0	1	0	0	1	0	0
1	1	1	1	1	1	1	1	1

Tautological Equivalence

Example

Show that $(p \rightarrow r) \wedge (q \rightarrow r) \Leftrightarrow (p \vee q) \rightarrow r$.

Tautological Equivalence

Example

Show that $(p \rightarrow r) \wedge (q \rightarrow r) \Leftrightarrow (p \vee q) \rightarrow r$.

Proof.

$$\begin{aligned}(p \rightarrow r) \wedge (q \rightarrow r) &\Leftrightarrow (\neg p \vee r) \wedge (\neg q \vee r) \\ &\Leftrightarrow (\neg p \wedge \neg q) \vee r \\ &\Leftrightarrow (\neg(p \vee q)) \vee r \\ &\Leftrightarrow (p \vee q) \rightarrow r\end{aligned}$$



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Example

Show that $(p \rightarrow r) \vee (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$.

Proof.

$$\begin{aligned}(p \rightarrow r) \vee (q \rightarrow r) &\Leftrightarrow (\neg p) \vee r \vee (\neg q) \vee r \\ &\Leftrightarrow (\neg p) \vee (\neg q) \vee r \\ &\Leftrightarrow (\neg(p \wedge q)) \vee r \\ &\Leftrightarrow (p \wedge q) \rightarrow r\end{aligned}$$



Definitions

A compound proposition is **satisfiable** if there is an assignment of truth values to its variables that makes it true (that is, when it is a tautology or a contingency).

If no such assignment exists (*i.e.* the proposition is false for all assignments of truth values to its variables), then the compound proposition is **unsatisfiable**. (Its negation is then a tautology: true for all truth assignments.)

A truth assignment that makes a compound proposition true is called a **solution of the satisfiability problem**.

Exercise

Determine whether the following compound propositions are satisfiable.

$$A = (p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p),$$

$$B = (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r),$$

$$C = (p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r).$$

Generalised De Morgan's Laws

We denote the n-ary AND and n-ary OR symbols by

$$\bigwedge_{j=1}^n p_j := p_1 \wedge p_2 \wedge \cdots \wedge p_n ;$$

$$\bigvee_{j=1}^n p_j := p_1 \vee p_2 \vee \cdots \vee p_n .$$

De Morgan's laws extend to

$$\neg \bigvee_{j=1}^n p_j = \bigwedge_{j=1}^n \neg p_j ;$$

$$\neg \bigwedge_{j=1}^n p_j = \bigvee_{j=1}^n \neg p_j .$$

Caution!

Don't apply logical identities, such as associative laws, distributive laws, or De Morgan's laws, to expressions that have a mix of conjunctions and disjunctions.

Exercise

Sudoku puzzles consist of a 9×9 grid made up of nine 3×3 subgrids known as blocks. Each block contains all 9 numbers $1, 2, \dots, 9$, each in one of its cells.

Let $p(i, j, n)$ be the proposition “cell in row i and column j has the value n ”.

1. Assert that row 1 of the puzzle contains all numbers from 1 to 9.
2. Assert that every row contains all numbers from 1 to 9.
3. Assert that every column contains all numbers from 1 to 9.
4. Assert that every block contains all numbers from 1 to 9.

1.2. Rules of Inference

Definitions

- ▶ Mathematical proofs are valid arguments that establish the truth of a mathematical statement.
- ▶ An **argument** in propositional logic is a sequence of propositions. All of those propositions are **premises**, except for the final one: the **conclusion**.
- ▶ An argument is **valid** if the truth of all its premises implies that the conclusion is true.
- ▶ An **argument form** in propositional logic is a sequence of compound propositions involving **propositional variables**.
- ▶ An argument form is **valid** if, no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if all premises are true.
- ▶ **Rules of inference** are some relatively simple argument forms whose validity we establish. They can be used as building blocks to construct more complicated valid argument forms.

Formal Implication

Definition

We say that p formally implies q if $p \rightarrow q$ is a tautology.

We denote it by $p \Rightarrow q$.

Example

Disjunctive addition:

$$p \Rightarrow (p \vee q)$$

p	q	$p \vee q$	$p \rightarrow p \vee q$
0	0	0	1
0	1	1	1
1	0	1	1
1	1	1	1

Summary (in terms of truth tables)

- ▶ Tautology: all ones in the truth table;
- ▶ Contradiction: all zeros in the truth table;
- ▶ Equivalence $p \Leftrightarrow q$: same value in the truth table;
- ▶ Formal Implication $p \Rightarrow q$: if $p = 1$, then $q = 1$.

Note that $p \Leftrightarrow q$ iff $p \Rightarrow q$ and $q \Rightarrow p$.

Example

Recall the the absorption rule $p \wedge (p \vee q) \Leftrightarrow p$. To establish this, we can show both $p \wedge (p \vee q) \Rightarrow p$ and $p \Rightarrow p \wedge (p \vee q)$.

- ▶ $p \wedge (p \vee q) \Rightarrow p$.

Indeed, since if $p \wedge (p \vee q) = 1$, then $p = 1$.

- ▶ $p \Rightarrow p \wedge (p \vee q)$.

Indeed, since if $p = 1$, then $p \wedge (p \vee q) = 1$ regardless of q .

Rules of Inference/Formal Implication/Deduction

- ▶ Detachment (Modus ponens): $(p \rightarrow q) \wedge p \Rightarrow q$
- ▶ Indirect reasoning (Modus tollens): $(p \rightarrow q) \wedge \neg q \Rightarrow \neg p$
- ▶ Disjunctive addition: $p \Rightarrow (p \vee q)$
- ▶ Conjunctive simplification: $(p \wedge q) \Rightarrow p$
- ▶ Disjunctive syllogism: $(p \vee q) \wedge \neg p \Rightarrow q$
- ▶ Hypothetical syllogism: $(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow p \rightarrow r$
- ▶ Resolution: $(p \vee q) \wedge (\neg p \vee r) \Rightarrow q \vee r$

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Proof by Contrapositive

$$p \rightarrow q \Leftrightarrow (\neg q \rightarrow \neg p)$$

Proof by Contradiction

$$p \rightarrow q \Leftrightarrow (p \wedge \neg q) \rightarrow 0 \Leftrightarrow \neg(p \wedge \neg q)$$

Note that $\neg p \Leftrightarrow p \rightarrow 0$.

An Example

Recall that for propositions A, B, C , we have

$$(A \wedge B) \rightarrow C \Leftrightarrow A \rightarrow B \rightarrow C \Leftrightarrow B \rightarrow A \rightarrow C$$

Given a prime number $p \in \mathbb{P}$, and $a, b \in \mathbb{N}$, let

- ▶ $A = p \mid ab$
- ▶ $B = p \nmid a = \neg p \mid a$
- ▶ $C = p \mid b$

$$p \mid ab \text{ iff } \exists q \in \mathbb{N} \text{ s.t. } pq = ab$$

Consider the following propositions

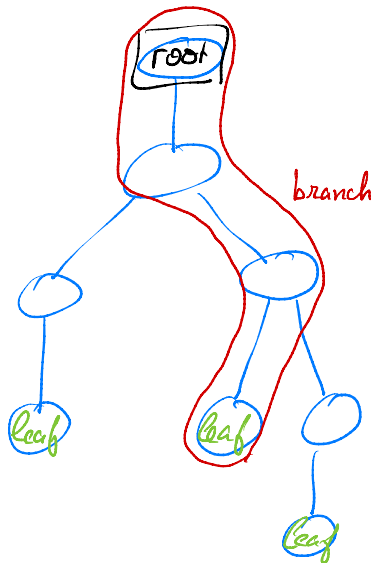
- ▶ $(p \mid ab) \wedge (p \nmid a) \rightarrow (p \mid b)$
- ▶ $(p \mid ab) \rightarrow (p \nmid a) \rightarrow (p \mid b)$
- ▶ $(p \mid ab) \rightarrow (p \mid a) \vee (p \mid b)$
- ▶ ...

Remark

Note that for $x, y \in \mathbb{N}$, we say x divides y , or y is divisible by x iff $\exists c \in \mathbb{N}, y = cx$, denoted $x \mid y$.

Method of Truth Trees/Semantic Tableau

- ▶ Method for proving or refuting a compound proposition.
- ▶ Can be used to determine satisfiability.
- ▶ Consists in iteratively constructing a tree following a set of rules.
- ▶ A **rooted tree** consists of **nodes** connected by **edges** that form no cycles. Each edge connects a **parent node**, above, to a **child node** below it. Each node has exactly one parent, except the top node, named the **root**. A childless node is named a **leaf** - at the bottom of the tree. A **branch** is a child-parent chain connecting a leaf to the root.



► At the start, proposition A occupies the root: the only node. It's an open node.

► Iterative method:

1. Pick an open leaf node.
2. Pick an applicable node on the branch above that leaf.
3. Expand the tree below the leaf by applying the applicable node based on some expansion rule.
4. For every newly created node that is both a literal/negated literal, and whose complement appears in a prior node on the same branch, mark the branch as closed. Mark all other newly created nodes as open.

$$\neg (p \rightarrow q) \Leftrightarrow \neg (\neg p \vee q) \Leftrightarrow p \wedge \neg q$$

$$(p \rightarrow q) \vee p \rightarrow q$$

$$(p \rightarrow q) \vee p$$

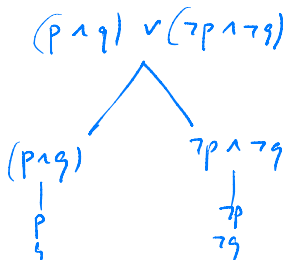
$$\neg q$$

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
1	1			
1	0			
0	1			
0	0			

$$(p \leftrightarrow q) \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$$

$$\Leftrightarrow (\neg p \vee q) \wedge (\neg q \vee p)$$

$$\Leftrightarrow (p \wedge q) \vee (\neg p \wedge \neg q)$$

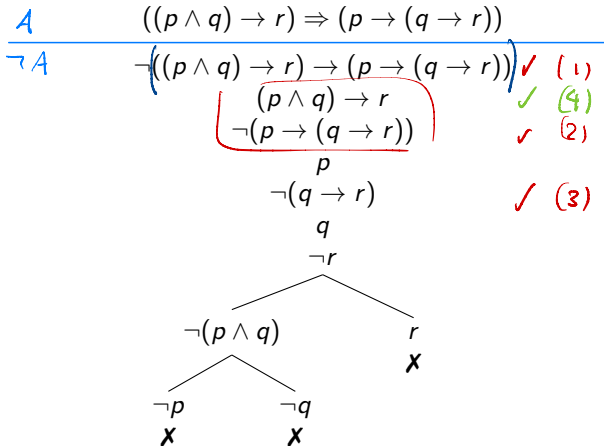


Method of Truth Trees/Semantic Tableau

$p \wedge q$ $\vdots$ p q	$\neg(p \vee q)$ $\vdots$ $\neg p$ $\neg q$	$\neg(p \rightarrow q)$ $\vdots$ p $\neg q$	$\neg\neg p$ $\vdots$ p
$\neg(p \wedge q)$ $\vdots$ $\neg p \quad \neg q$	$p \vee q$ $\vdots$ $p \quad q$	$p \rightarrow q$ $\vdots$ $\neg p \quad q$	
$p \leftrightarrow q$ $\vdots$ $p \quad \neg p$ $q \quad \neg q$	$\neg(p \leftrightarrow q)$ $\vdots$ $p \quad \neg p$ $\neg q \quad q$		

Method of Truth Trees - Example 1

We can prove $A := ((p \wedge q) \rightarrow r) \Rightarrow (p \rightarrow (q \rightarrow r))$ using the following truth tree method (proof by contradiction):



- ▶ We write $\neg A$ at the root.
- ▶ We expand the tree applying the rules.
- ▶ Recall: a branch is **closed** when it contains a proposition and its negation both appear on it.
- ▶ If all branches are closed, then $\neg A$ is a contradiction. Hence A is a tautology.

$\neg A := \neg((p \wedge q) \rightarrow r) \rightarrow (p \rightarrow (q \rightarrow r))$ root
leaf

$$\neg((p \wedge q) \rightarrow r) \rightarrow (p \rightarrow (q \rightarrow r))$$

root
leaf

(1)

has been applied -

(1) $(p \wedge q) \rightarrow r$

can be applied

(1) $\neg(p \rightarrow (q \rightarrow r))$

can be applied

leaf

$$\neg((p \wedge q) \rightarrow r) \rightarrow (p \rightarrow (q \rightarrow r))$$

root
leaf

(1)

has been applied -

(1) $(p \wedge q) \rightarrow r$

can be applied

(1) $\neg(p \rightarrow (q \rightarrow r))$

leaf

(2)

~~can be~~ applied
has been

(2) p

cannot be applied:
it's a literal -

(2) $\neg(q \rightarrow r)$

leaf

can be applied.

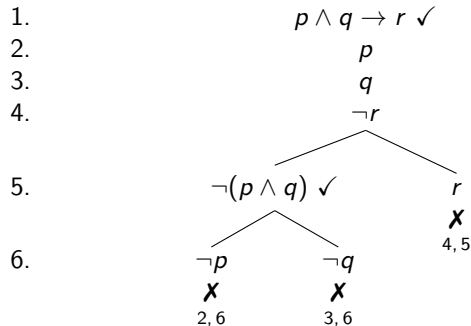
Method of Truth Trees - Example 2

We can also write $((p \wedge q) \rightarrow r) \Rightarrow (p \rightarrow (q \rightarrow r))$ in the *sequent* form $p \wedge q \rightarrow r \vdash p \rightarrow q \rightarrow r$, where the sequent $p_1, \dots, p_n \vdash q_1, \dots, q_m$ means $p_1 \wedge \dots \wedge p_n \Rightarrow q_1 \vee \dots \vee q_m$.

$\vdash$: the symbol is a turnstile
read as "entails" (syntactically) or "proves".

Again by contradiction: input the premise $p \wedge q \rightarrow r$ and a negation of the consequent at the root:

$$p \wedge q \rightarrow r \vdash p \rightarrow q \rightarrow r$$



premise
antecedent
antecedent
 $\neg$ consequent

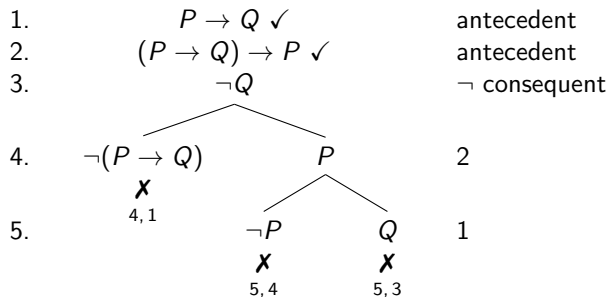
$$\begin{aligned} & \neg(p \rightarrow q \rightarrow r) \\ \Leftrightarrow & \neg(p \rightarrow (q \rightarrow r)) \\ \Leftrightarrow & \neg(\neg p \vee (q \rightarrow r)) \\ \Leftrightarrow & \neg\neg p \wedge \neg(q \rightarrow r) \\ \Leftrightarrow & p \wedge \neg(\neg q \vee r) \\ \Leftrightarrow & p \wedge q \wedge \neg r. \end{aligned}$$

Method of Truth Trees - Example 3

Example

We can close a branch whenever a contradiction is detected.

$$\vdash (P \rightarrow Q) \rightarrow ((P \rightarrow Q) \rightarrow P) \rightarrow Q$$



cf., [https://www.umsu.de/trees/#\(P~5Q\),\(\(P~5Q\)~5P\)%7C=Q](https://www.umsu.de/trees/#(P~5Q),((P~5Q)~5P)%7C=Q)

$$(\neg q \rightarrow \neg p) \wedge (\neg r \rightarrow \neg q), s \wedge (s \rightarrow \neg r), t \rightarrow p \vdash \neg t$$

1.	$(\neg q \rightarrow \neg p) \wedge (\neg r \rightarrow \neg q) \checkmark$	premise
2.	$s \wedge (s \rightarrow \neg r) \checkmark$	premise
3.	$t \rightarrow p$	premise
4.	$\neg \neg t$	$\neg$ consequent
5.	s	2
6.	$s \rightarrow \neg r \checkmark$	2
7.	$\neg q \rightarrow \neg p$	1
8.	$\neg r \rightarrow \neg q \checkmark$	1
9.	$\neg s$ $\mathbf{X}$ 5, 9	6
10.	$\neg \neg r$ $\mathbf{X}$ 9, 10	8
11.	$\neg \neg q$ $\mathbf{X}$ 10, 11	7
12.	$\neg t$ $\mathbf{X}$ 4, 12	3
	p $\mathbf{X}$ 11, 12	

1.3. Predicates and Quantifiers

Statements

Propositional logic cannot express all statements we may want to.

Examples

- ▶ “ $5 > 3$ ” is a true proposition.

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- ▶ “ $5 > 3$ ” is a **true proposition**.
- ▶ “Every natural number is greater than 3” is a **false proposition**.
- ▶ “ $x > 3$ ” is **not a proposition**, because we can not decide whether it is true or not. It is a statement form composed of 2 parts:
 - ▶ A **propositional variable** x , subject of
 - ▶ a **predicate**, or **propositional function** P ,
where $P(x)$ denotes the statement form “ $x > 3$ ”.

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A propositional function P with n variables $x_1, \dots, x_n$ is called an **n -ary predicate**. P can be evaluated at the **n -tuple** $(x_1, \dots, x_n)$ as $P(x_1, \dots, x_n)$.

Statements

Definition

A **predicate** is a function $P : X \rightarrow \{\top, \perp\}$ with domain (or “universe”) X .

Remark

A statement form or predicate becomes a statement (which can then be either true or false) when the variable takes on a specific value.

Example

If $P(x)$ stands for “ $x > 3$ ”, then

- ▶ $P(4) = \top$, i.e., $4 > 3$ is a TRUE statement;
- ▶ $P(2) = \perp$, i.e., $2 > 3$ is a FALSE statement.

We indicate that an *element* x is a member of a *set* A by writing $x \in A$. We may characterize the elements of a set A by some predicate P , e.g.,

in **set-builder notation**:

$$A = \{x : P(x)\} = \{x \mid P(x)\}.$$

Hence:

$$x \in \{y \mid P(y)\} \Leftrightarrow P(x).$$

Definition (Sets)

- ▶ A **set** A is an unordered collection of distinct **elements**. A is said to **contain** its elements. We denote by
 - ▶ $a \in A$ the proposition “ a is an element of A ”, and by
 - ▶ $a \notin A$ the proposition “ a is not an element of A ”.
- ▶ A is a **subset** of a set B , denoted by $A \subseteq B$, iff every element of A is also an element of B . $A \subset B$ $A \not\subseteq B$ iff $A \subseteq B$ and $A \neq B$
- ▶ Such a set B is called a **superset** of A , denoted by $B \supseteq A$.

Definition (Set-builder notation)

A set A can be described:

- ▶ by enumerating all its elements: $A = \{x_1, x_2, \dots\}$,
- ▶ or in **set-builder notation**: $A = \{x \mid P(x)\}$:
“ A is the set of all objects x that satisfy proposition $P(x)$ ”.

Quick Reminder - Set Operations

Definition (Set Operations)

Let A, B be two sets.

- ▶ The **intersection** $A \cap B$ is the set of all objects x that are elements of both sets A and B :

$$A \cap B = \{x \mid (x \in A) \wedge (x \in B)\}.$$

- ▶ The **union** $A \cup B$ is the set of all objects x that are elements of at least one of the sets A and B :

$$A \cup B = \{x \mid (x \in A) \vee (x \in B)\}.$$

We denote by $\emptyset = \{\}$ the **empty set**, of the set without any element. For any set A , we have: $A \cap \emptyset = \emptyset$, and $A \cup \emptyset = A$.

Logical Quantifiers

There are two types of quantifiers (over some “universe” U):

- ▶ the **universal quantifier**, denoted by the symbol $\forall$, read as “for all” and

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Definition (Universal Quantifier)

$$P(\cdot, x_2, x_3) : x \mapsto P(x, x_2, x_3)$$

Let U be a set (the “universe”) and $P(\cdot)$ a predicate. We define the universal quantifier $\forall$ by

$$\begin{aligned}\forall x \in U, P(x) &\Leftrightarrow \text{for all } x \in U, P(x) \text{ is true} &&\Leftrightarrow \text{For all } x, \text{ if } x \in U, \text{ then } P(x) \\ &\Leftrightarrow \forall x, x \in U \rightarrow P(x) &&\Leftrightarrow \{x \mid x \in U\} \subseteq \{x \mid P(x)\} \\ &\Leftrightarrow U \subseteq \{x \mid P(x)\}\end{aligned}$$

where $U = \{x \mid P(x)\}$ if U is the universe.

Definition (Existential Quantifier)

Let U be a set (the “universe”) and $P(\cdot)$ a predicate. We define the existential quantifier $\exists$ by

$\exists x \in U, P(x) \Leftrightarrow$ There is *at least one* $x \in U$ such that $P(x)$ is true.

$$\Leftrightarrow \{x \mid x \in U\} \cap \{x \mid P(x)\} \neq \emptyset$$

$$\Leftrightarrow \{x \mid x \in U \wedge P(x)\} \neq \emptyset$$

$$\Leftrightarrow \{x \in U \mid P(x)\} \neq \emptyset$$

Other forms of quantification

We may also write $\forall x \in U: P(x)$, $\forall_{x \in U} P(x)$, or $\forall_{x \in U} P(x)$ instead of $(\forall x \in U)P(x)$. Similarly for $\exists$.

Restricted/Unrestricted quantifiers

Note that

$$\blacktriangleright \forall x \in U, P(x) \Leftrightarrow \forall x, x \in U \rightarrow P(x)$$

$$\blacktriangleright \exists x \in U, P(x) \Leftrightarrow \exists x, x \in U \wedge P(x)$$

Example

$$\blacktriangleright \forall x > 0, x^2 > 0 \Leftrightarrow \forall x, x > 0 \rightarrow x^2 > 0$$

$$\blacktriangleright \exists z > 0, z^2 = 2 \Leftrightarrow \exists z, z > 0 \wedge z^2 = 2$$

It is also common to represent “there exists a unique x such that $P(x)$ is true” as $\exists! x P(x)$, meaning

$$\exists x(P(x) \wedge \forall y(P(y) \rightarrow y = x)) \quad \text{or} \quad \exists x \forall y(P(y) \leftrightarrow y = x)$$

Scope and Order of Priority of Quantifiers

- ▶ Quantifiers have higher precedence than all logical operators from propositional calculus:

$$\forall x P(x) \vee Q(x) \Leftrightarrow (\forall x P(x)) \vee Q(x).$$

- ▶ The **scope** of a quantifier is the shortest formula in which it occurs, determining the range in the formula to which the quantifier is applied.
- ▶ A variable that is within the range of a quantifier is a **bound variable**,
- ▶ A variable that is outside the range of all quantifiers is a **free variable**.

Exercise

Determine the free and bound variables in the expressions below:

The diagram shows two logical expressions with handwritten annotations for free and bound variables. In the first expression, $\forall x (P(x) \rightarrow Q(x, y)) \rightarrow \exists y (R(x, y) \vee \forall x \neg S(x))$, the scope of $\forall x$ is $(P(x) \rightarrow Q(x, y))$ (labeled 'bound'), the scope of $\exists y$ is $(R(x, y) \vee \forall x \neg S(x))$ (labeled 'bound'), and the x in $R(x, y)$ is labeled 'free'. In the second expression, $\exists x \forall y (P(x) \vee Q(x, y)) \rightarrow R(x, y) \rightarrow \forall y S(y)$, the scope of $\exists x$ is $(P(x) \vee Q(x, y))$ (labeled 'bound'), the scope of $\forall y$ is $(P(x) \vee Q(x, y))$ (labeled 'bound'), and both the x and y in $R(x, y)$ are labeled 'free'.

$$\forall x (P(x) \rightarrow Q(x, y)) \rightarrow \exists y (R(x, y) \vee \forall x \neg S(x)),$$
$$\exists x \forall y (P(x) \vee Q(x, y)) \rightarrow R(x, y) \rightarrow \forall y S(y).$$

Remark: A proposition has no free variables.

We may also state the domain/universe before making the statements.

Examples

Let x be a real number. Then

- ▶ $\forall x, x > 0 \rightarrow x^3 > 0$ is a true statement;

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Logical Quantifiers

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Let x be a real number. Then

- ▶ $\forall x, x > 0 \rightarrow x^3 > 0$ is a true statement;
- ▶ $\forall x, x > 0 \leftrightarrow x^2 > 0$ is a false statement;
- ▶ $\exists x, x > 0 \leftrightarrow x^2 > 0$ is a true statement.

Hanging Quantifier

Sometimes quantifiers are put at the end (if no ambiguities arise)

$$\exists y: y + x^2 > 0 \qquad \forall x$$

is equivalent to $\exists y \forall x: y + x^2 > 0$.

Contraposition and Negation of Quantifiers

We can represent the quantifier $\exists$ using $\neg$ and $\forall$, semi-formally,

$$\begin{aligned}\exists x \in U, P(x) &\Leftrightarrow \{x \in U \mid P(x)\} \neq \emptyset \\ &\Leftrightarrow \neg[\{x \in U \mid P(x)\} = \emptyset] \\ &\Leftrightarrow \neg[\{x \in U \mid \neg P(x)\} = U] \\ &\Leftrightarrow \neg[\forall x \in U, \neg P(x)]\end{aligned}$$

$$\neg(a=b) \Leftrightarrow a \neq b$$

The last equivalence is called *contraposition of quantifiers*. It implies that

$$\neg[\exists x \in U, P(x)] \Leftrightarrow \forall x \in U, \neg P(x)$$

For example,

$$\neg(\exists x \in \mathbb{R}, x^2 < 0) \Leftrightarrow \forall x \in \mathbb{R}, x^2 \not< 0$$

Conversely,

$$\neg[\forall x \in U, P(x)] \Leftrightarrow \exists x \in U, \neg P(x)$$

De Morgan's Laws for Quantifiers

It is intuitive to consider the following

$$\bigwedge_{i=1}^n P(x_i)$$

$$\forall x \in \{x_1, x_2, \dots, x_n\}, P(x) \Leftrightarrow P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$$

$$\exists x \in \{x_1, x_2, \dots, x_n\}, P(x) \Leftrightarrow P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$$

thus

$$\neg[\forall x \in \{x_1, x_2, \dots, x_n\}, P(x)] \Leftrightarrow \neg P(x_1) \vee \neg P(x_2) \vee \dots \vee \neg P(x_n)$$

$$\neg[\exists x \in \{x_1, x_2, \dots, x_n\}, P(x)] \Leftrightarrow \neg P(x_1) \wedge \neg P(x_2) \wedge \dots \wedge \neg P(x_n)$$

To summarize,

- ▶ $\neg[\exists x \in U, P(x)] \Leftrightarrow \forall x \in U, \neg P(x)$
- ▶ $\neg[\forall x \in U, P(x)] \Leftrightarrow \exists x \in U, \neg P(x)$
- ▶ $\neg[\forall x \in U, \neg P(x)] \Leftrightarrow \exists x \in U, P(x)$
- ▶ $\neg[\exists x \in U, \neg P(x)] \Leftrightarrow \forall x \in U, P(x)$

(Meta-)Theorems in Predicate Logic/First-order Logic

- $\vdash \forall x \in U, P(x) \vee \neg P(x)$
- $\neg[\forall x \in U, P(x)] \dashv\vdash \exists x \in U, \neg P(x)$
- $\neg[\exists x \in U, P(x)] \dashv\vdash \forall x \in U, \neg P(x)$
- $\forall x \in U, P(x) \vdash \exists x \in U, P(x)$, if $U \neq \emptyset$
- $\exists x \in U, [P(x) \vee Q(x)] \dashv\vdash [\exists x \in U, P(x)] \vee [\exists x \in U, Q(x)]$
- $\forall x \in U, [P(x) \wedge Q(x)] \dashv\vdash [\forall x \in U, P(x)] \wedge [\forall x \in U, Q(x)]$
- $\exists x \in U, [P(x) \wedge Q(x)] \vdash [\exists x \in U, P(x)] \wedge [\exists x \in U, Q(x)]$
- $\forall x \in U, [P(x) \vee Q(x)] \vdash [\forall x \in U, P(x)] \vee [\forall x \in U, Q(x)]$
- $\exists x \in U, [P(x) \wedge Q(x)] \vdash [\forall x \in U, P(x)] \wedge [\exists x \in U, Q(x)]$
- $[\forall x \in U, [P(x) \rightarrow Q(x)]] \wedge [\forall x \in U, P(x)] \vdash \forall x \in U, Q(x)$
- $\forall x \in \{y \in U \mid P(y)\}, Q(x) \dashv\vdash \forall x \in U, [P(x) \rightarrow Q(x)]$
- $\exists x \in \{y \in U \mid P(y)\}, Q(x) \dashv\vdash \exists x \in U, [P(x) \wedge Q(x)]$
- $\varphi \wedge [\exists x \in U, P(x)] \dashv\vdash \exists x \in U, [\varphi \wedge P(x)]$, if no x in φ
- $\varphi \vee [\forall x \in U, P(x)] \dashv\vdash \forall x \in U, [\varphi \vee P(x)]$, if no x in φ
- $\vdash$ turnstile : "proves"
- $\vdash \forall x \in \emptyset, P(x)$
- $\vdash \neg \exists x \in \emptyset, P(x)$

Recall the equivalence

$$\forall x \in U, P(x) \Leftrightarrow \forall x, [x \in U \rightarrow P(x)]$$

If the domain of the universal quantifier $\forall$ is the empty set $U = \emptyset$, then the statement

$$\forall x \in \emptyset, P(x)$$

is true regardless of the predicate P . It is then said that $P(x)$ is *vacuously true*.

Vacuous Truth

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Let $U := \{x \in \mathbb{R} \mid x \neq x\}$. Then the statement $(\forall x \in U)(x > x)$ is true.

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Example

Let $U := \{x \in \mathbb{R} \mid x \neq x\}$. Then the statement $(\forall x \in U)(x > x)$ is true.

This is similar to saying that “all pink elephants can fly” is a true statement, because it is impossible (as far as we know) to find a pink elephant that can or cannot fly.

Nested Quantifiers

We can also treat predicates with more than one variable. Quantifiers can be grouped into blocks as

$$\forall a \forall b \dots \forall c \quad \exists u \exists v \dots \exists w \quad \forall x \forall y \dots \forall z$$

In general, quantifiers can be swapped inside a block, but not between blocks.

Examples

In the following examples, $x, y \in \mathbb{R}$.

- $\forall x \forall y: x^2 + y^2 - 2xy \geq 0$ is equivalent to $\forall y \forall x: x^2 + y^2 - 2xy \geq 0$.
Therefore, one often writes $\forall x, y: x^2 + y^2 - 2xy \geq 0$.

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- ▶ $\exists x \exists y: x + y > 0$ is equivalent to $\exists y \exists x: x + y > 0$, often abbreviated to $\exists x, y: x + y > 0$.

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Therefore, one often writes $\forall x, y: x^2 + y^2 - 2xy \geq 0$.
- ▶ $\exists x \exists y: x + y > 0$ is equivalent to $\exists y \exists x: x + y > 0$, often abbreviated to $\exists x, y: x + y > 0$.
- ▶ $\forall x \exists y: x + y > 0$ is a true statement. (y depends on x)
- ▶ $\exists y \forall x: x + y > 0$ is a false statement. (x does not depend on y)

Nested Quantifiers

Consider $M = \{1, 2, 3, 4, 5\}$, and the predicate

$$P : M \times M \rightarrow \{\top, \perp\}, \quad (x, y) \mapsto P(x, y).$$

$P(x, y)$	1	2	3	4	5	
1	$\top$	$\perp$	$\perp$	$\top$	$\top$	(i) $\vdash \forall x \exists y P(x, y)$
2	$\top$	$\perp$	$\top$	$\top$	$\top$	(ii) $\vdash \exists y \forall x P(x, y)$
3	$\perp$	$\perp$	$\top$	$\top$	$\top$	(iii) $\vdash \forall y \exists x P(x, y)$
4	$\perp$	$\top$	$\perp$	$\top$	$\perp$	
5	$\top$	$\perp$	$\top$	$\top$	$\top$	(iv) $\not\vdash \exists x \forall y P(x, y)$

Note that we can get (i) from (ii) for free, since

$$\exists y \forall x P(x, y) \vdash \forall x \exists y P(x, y)$$

which can be viewed as a full-blown version of

$$\exists y (P_{x_1}(y) \wedge P_{x_2}(y)) \vdash \exists y P_{x_1}(y) \wedge \exists y P_{x_2}(y)$$

Nested Quantifiers

Example: $f_n \rightarrow f$

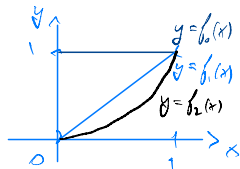
- Pointwise convergence: $\lim_{n \rightarrow \infty} (f_n(x) - f(x)) = 0$ for all $x \in I$

$$\forall \varepsilon > 0, \forall x \in I, \exists n \in \mathbb{N}, \forall m \geq n, \|f_m(x) - f(x)\| < \varepsilon$$

- Uniform convergence: $\lim_{n \rightarrow \infty} \sup_{x \in I} \|f_n(x) - f(x)\| = 0$

$$\forall \varepsilon > 0, \exists n \in \mathbb{N}, \forall x \in I, \forall m \geq n, \|f_m(x) - f(x)\| < \varepsilon$$

$$f_n : (0, 1) \longrightarrow \mathbb{R}$$
$$x \longmapsto x^n$$



1.4. Introduction to Formal Proofs

Formal and Informal Proofs

- ▶ A *proof* is some kind of *deduction* (*derivation*) that proceeds from a set of *hypotheses* (*premises*, *axioms*) in order to derive a *conclusion*, using some *logical rules*.
- ▶ Proofs can be very informal, using a set of loosely defined logical rules, possibly omitting steps and premises.
- ▶ Proofs can be completely formal, using a very clearly defined set of rules and premises. Such proofs are usually processed or produced by programs called proof checkers and theorem provers (LEAN, COQ, *etc.*).
- ▶ Thus, a human prover evolves in a spectrum of formality:
 - ▶ It is extremely time- (and space-)consuming to write or read fully formal proofs: it is rarely the best choice.
 - ▶ But there are critical systems (e.g. flight-control, nuclear reactor, ...) in which we must have absolute confidence.

Axiomatic Set Theory Examples

- ▶ Extensionality Axiom

$$\forall A, B (\forall x (x \in A \leftrightarrow x \in B) \rightarrow A = B)$$

- ▶ Empty Set Axiom

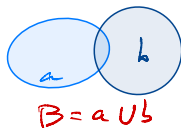
$$\exists B \forall x (x \notin B)$$

- ▶ Pairing Axiom

$$\forall u, v \exists B \forall x (x \in B \leftrightarrow x = u \vee x = v)$$

- ▶ Union Axiom

$$\forall a, b \exists B \forall x (x \in B \leftrightarrow x \in a \vee x \in b)$$



- ▶ Powerset Axiom

$$\forall a \exists b \forall x (x \in b \leftrightarrow x \subseteq a)$$

where $x \subseteq a$ is shorthand for $\forall t (t \in x \rightarrow t \in a)$. *b is the set of all subsets of a*

$$a = \{0, 1\}$$

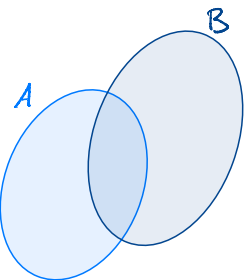
$$\emptyset \subseteq a; \quad \{0\} \subseteq a; \quad \{1\} \subseteq a$$

$$\{0, 1\} \subseteq a$$

$$b = \{\emptyset, \{0\}, \{1\}, a\}$$

Set-Theoretic Proofs

Basic Facts



$$x \in A \cup B \Leftrightarrow (x \in A) \vee (x \in B)$$

$$x \notin A \cup B \Leftrightarrow (x \notin A) \wedge (x \notin B)$$

$$x \in A \cap B \Leftrightarrow (x \in A) \wedge (x \in B)$$

$$x \notin A \cap B \Leftrightarrow (x \notin A) \vee (x \notin B)$$

$$x \in A - B \Leftrightarrow (x \in A) \wedge (x \notin B)$$

$$x \notin A - B \Leftrightarrow (x \notin A) \vee (x \in B)$$

$$A \subseteq B \Leftrightarrow \forall x, (x \in A \rightarrow x \in B)$$

$$A - B = A \setminus B = \{x \in A \mid x \notin B\}$$

Basic Facts

$$x \in A \cup B \iff (x \in A) \vee (x \in B)$$

$$x \notin A \cup B \iff (x \notin A) \wedge (x \notin B)$$

$$x \in A \cap B \iff (x \in A) \wedge (x \in B)$$

$$x \notin A \cap B \iff (x \notin A) \vee (x \notin B)$$

$$x \in A - B \iff (x \in A) \wedge (x \notin B)$$

$$x \notin A - B \iff (x \notin A) \vee (x \in B)$$

$$A \subseteq B \iff \forall x, (x \in A \rightarrow x \in B)$$

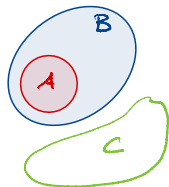
Remark

- ▶ Prove something exists: sufficient to find an example.
- ▶ Prove $P(x)$ for all $x \in A$: Take any $x \in A$ and continue. (or use induction if $A = \mathbb{N}$, more on this later.)

Set-Theoretic Proofs

Example

Given sets A, B, C , if $A \subseteq B$ and $B \cap C = \emptyset$, then $A \cap C = \emptyset$.



Proof.

Note that in first-order logic, the statement is

$$\forall x(x \in A \rightarrow x \in B), \forall x \neg(x \in B \wedge x \in C) \vdash \forall x \neg(x \in A \wedge x \in C)$$

Let x be an arbitrary element, it suffices to show the following proposition

$$x \in A \rightarrow x \in B, \neg(x \in B \wedge x \in C) \vdash \neg(x \in A \wedge x \in C)$$

Let p, q, r be the logical variables for $x \in A, x \in B, x \in C$, resp., we need to show

$$p \rightarrow q, \neg(q \wedge r) \vdash \neg(p \wedge r)$$

(which is clearly true since it is equivalent to $p \rightarrow q, p \wedge r \vdash q \wedge r$)

$$((p \rightarrow q) \wedge \neg(q \wedge r)) \longrightarrow \neg(p \wedge r)$$

$$(A \wedge \neg B) \rightarrow \neg C$$

$$\Leftrightarrow A \rightarrow \neg B \rightarrow \neg C$$

$$\Leftrightarrow A \rightarrow (\neg B \rightarrow \neg C)$$

$$\Leftrightarrow A \rightarrow (C \rightarrow B)$$

$$\Leftrightarrow A \rightarrow C \rightarrow B$$

$$\Leftrightarrow (A \wedge C) \rightarrow B$$

Proof.

We can use truth table.

p	q	r	$p \rightarrow q$	$\neg(q \wedge r)$	$\neg(p \wedge r)$	$[(p \rightarrow q) \wedge \neg(q \wedge r)] \rightarrow \neg(p \wedge r)$
0	0	0	1	1	1	1
0	0	1	1	1	1	1
0	1	0	1	1	1	1
0	1	1	1	0	1	
1	0	0	0	1	1	
1	0	1	0	1	0	
1	1	0	1	1	1	1
1	1	1	1	0	0	

Set-Theoretic Proofs

Or use truth trees.

$$p \rightarrow q, \neg(q \wedge r) \vdash \neg(p \wedge r)$$

1.	$p \rightarrow q \checkmark$	premise
2.	$\neg(q \wedge r) \checkmark$	premise
3.	$p \wedge r \checkmark$	$\neg$ consequent
4.	p	3
5.	r	3
	$\swarrow$ $\searrow$	
6.	$\neg p$ X 4, 6	1
	$\swarrow$ $\searrow$	
7.	$\neg q$ X 6, 7	2
	$\swarrow$ $\searrow$	
	$\neg r$ X 5, 7	

[https://www.umsu.de/trees/#p~5q,~3\(q~1r\)%7C=~3\(p~1r\)](https://www.umsu.de/trees/#p~5q,~3(q~1r)%7C=~3(p~1r))

Set-Theoretic Proofs

Or use truth trees for the original sequent in first-order logic.

($Ax := x \in A$ etc.)

$$\forall x(Ax \rightarrow Bx), \forall x\neg(Bx \wedge Cx) \vdash \forall x\neg(Ax \wedge Cx)$$

1.	$\forall x(Ax \rightarrow Bx)$ ✖	premise
2.	$\forall x\neg(Bx \wedge Cx)$ ✖	premise
3.	$\exists x(Ax \wedge Cx)$ ✔	$\neg$ consequent
4.	$At \wedge Ct$ ✔	3 x/t
5.	At	4
6.	Ct	4
7.	$\neg(Bt \wedge Ct)$ ✔	2 x/t
8.	$At \rightarrow Bt$ ✔	1 x/t
9.	$\neg At$ ✖ 9, 5 Bt	8
10.	$\neg Bt$ ✖ 10, 9 $\neg Ct$ ✖ 10, 6	

Truth Trees for First-Order Logic

$\exists y \forall x Pxy \vdash \forall x \exists y Pxy$

- | | |
|----|----------------------------------|
| 1. | $\exists y \forall x Pxy$ ✓ |
| 2. | $\exists x \forall y \neg Pxy$ ✓ |
| 3. | $\forall y \neg Pay$ * |
| 4. | $\forall x Pxb$ * |
| 5. | $\neg Pab$ |
| 6. | Pab |
| | X |
| | 5, 6 |

premise $\Rightarrow \exists x \neg \exists y Pxy$
 $\neg$ consequent

- | | | |
|---|-------|--|
| 2 | x/a | $\Rightarrow \exists x \forall y \neg Pxy$ |
| 1 | y/b | |
| 3 | | |
| 4 | | |

<https://www.umsu.de/trees/#~7y~6xPxy%7C=~6x~7yPxy>

Strategies

- ▶ Push negation past all quantifiers, but no further
- ▶ Introduce new variables for $\exists$ (if applicable)
- ▶ Introduce existing variables for $\forall$ (if applicable)
- ▶ Can reuse $\forall$ -sentences

Truth Trees for First-Order Logic

$\exists x(\forall y(Py \rightarrow x = y) \wedge Px) \vdash \exists x\forall y(Py \leftrightarrow x = y)$

- | | | |
|----|---|-------------------|
| 1. | $\exists x(\forall y(Py \rightarrow x = y) \wedge Px) \checkmark$ | premise |
| 2. | $\forall x\exists y\neg(Py \leftrightarrow x = y) *$ | $\neg$ consequent |
| 3. | $\forall y(Py \rightarrow a = y) \wedge Pa \checkmark$ | 1 |
| 4. | $\forall y(Py \rightarrow a = y) *$ | 3 |
| 5. | Pa | 3 |
| 6. | $\exists y\neg(Py \leftrightarrow a = y) \checkmark$ | 2 |
| 7. | $\neg(Pb \leftrightarrow a = b) \checkmark$ | 6 |

